Agentic AI in Modern Business

2025 Winter Semester

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1 Introduction

Today, AI agents are beginning to act as digital assistants that can do more than simply answer questions. They can find information in emails, draft and send replies, schedule appointments, compare products under given requirements, plan a journey, write and run code, and carry out tasks across different applications. How can an AI agent understand what we want, decide what to do next, and use digital tools to complete a task?

In this intensive block-week course, we will take a journey through the foundations and development of agentic AI: from understanding how language models work to exploring context, memory, knowledge, tools, coding agents, evaluation, and reinforcement learning. Through hands-on exercises and a group project, you will build an AI agent that interacts with real digital environments, examine how it makes decisions, identify where it fails, and improve its performance.

2 Schedule

- Time: Everyday, **9:00AM-18:00PM**, from 21 Sept 2026 to 25 Sept 2026.
- Location: B.0.23, Seminarraum (1910.EG.023B)

3 Course Evaluation

- **Daily Work** [Individual, 40%] From Monday to Thursday, students will complete daily assignments related to the topics and practical exercises covered that day. Each day will conclude with a brief individual oral assessment based on randomly selected questions from the corresponding question bank. Assessment will consider conceptual understanding, the ability to explain key ideas clearly, and the successful completion of the assigned practical work.

- **AI Agent Project** [Group, 40%] The group project will be announced on Monday and completed in groups of 2–4 students throughout the week. Each group will select a course-inspired use case and develop an AI agent that performs a series of tasks in a digital environment. The final submission must include a working implementation, evaluation results, trajectory logs, an analysis of at least one failure, and one documented improvement supported by before-and-after results. Assessment will consider the project's design, implementation, evaluation quality, critical analysis, and reproducibility.

- **Presentation** [Group, 20%] On Friday afternoon, each group will give a 10-30 minute presentation followed by a 5-10 minute Q&A session. The presentation should explain the selected use case, agent design, implementation, evaluation results, key findings, and at least one unsuccessful attempt or failure revealed by the evaluation. After each presentation, students from all other groups are strongly encouraged to ask questions, share observations, and contribute to the discussion. Assessment will consider the clarity and structure of the presentation, the quality of the technical explanation and analysis, and its active participation in the discussions following other groups' presentations.

4 Topic Agenda

Date	Topic	Time	Topic
Monday	Agent Fundamentals	9:00AM-12:00AM	- Introduction to Large Language Models: What a language model actually computes - Agent = LLM + Context + Tools
		14:00PM-18:00PM	- Daily work assignment - Oral assessment
Tuesday	Context Engineering	9:00AM-12:00AM	- Prompt engineering and skills - Memory and the knowledge base
		14:00PM-18:00PM	- Daily work assignment / Group project - Oral assessment
Wednesday	Tools, Function Calling & the Action Space	9:00AM-12:00AM	- Tools are the action space - Tool design and the ecosystem
		14:00PM-18:00PM	- Daily work assignment / Group project - Oral assessment
Thursday	Coding Agents, Vibe Coding & Evaluation	9:00AM-12:00AM	- Code is the meta-capability of agents - Interaction with coding agents: vibe coding - Anatomy of an evaluation task
		14:00PM-18:00PM	- Group project
Friday	Reinforcement Learning	9:00AM-12:00AM	- From pre-training to RL - How to train an agent with RL
		14:00PM-18:00PM	- Group project - Group project presentation

5 Interested in going deeper in this topic?

If you are interested in doing some research with our chair in the broad topics of machine learning and data mining, feel free to reach us out for a coffee or a small conversation. For more details about potential research, please check the website of Prof. Ziyue Li: https://bonaldli.github.io/